Joint-Aware KL Classification for Hand Osteoarthritis Using PIP Joint Radiographs

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Abstract

This project investigates whether proximal interphalangeal (PIP) joints (PIP2–PIP5) can be treated interchangeably in automated Kellgren–Lawrence (KL) classification, or whether joint-specific differences must be explicitly modeled. We design a comprehensive experimental pipeline combining convolutional neural networks (CNNs), joint classification tasks, feature visualization, and statistical testing.

Our results demonstrate that PIP joints exhibit strong joint-specific characteristics. Incorporating joint position as an additional input improves macro F1 score from 0.4666 to 0.5864, representing a significant performance gain. Statistical analysis further confirms highly significant differences across joints ($p \approx 0$). We conclude that joint-aware modeling is essential for accurate KL classification.

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1. Introduction

Osteoarthritis severity is commonly assessed using the Kellgren–Lawrence (KL) grading system, which evaluates radiographic joint degeneration. While deep learning methods have been widely applied to automate KL classification, an important question remains:

Can different PIP joints (PIP2–PIP5) be treated as identical inputs, or do they exhibit joint-specific patterns?

This study aims to answer this question through a combination of:

- Image-based classification models
- Joint discrimination tasks
- Statistical feature analysis

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2. Dataset and Preprocessing

2.1 Dataset Overview

- Total usable images: **13,772**
- Source: Hand joint radiographs

- Labels: Extracted from structured metadata file

2.2 KL Distribution

KL Grade	Count
0	9776
1	2217
2	1665
3	69
4	45

The dataset exhibits **severe class imbalance**, particularly for KL3 and KL4.

2.3 Joint Distribution

Joint	Count
PIP2	3455
PIP3	3430
PIP4	3447
PIP5	3440

2.4 Data Splitting

- Train: 9632 images
- Validation: 2068 images
- Test: 2072 images

A **patient-level split** is used to prevent data leakage.

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3. Methodology

3.1 Model Architecture

We use a **ResNet18-based convolutional neural network**:

- Input: grayscale image
- Output: KL grade (0–4)

3.2 Joint Index Encoding

Each joint is encoded as a numerical index:

Joint	Index
PIP2	0
PIP3	1
PIP4	2
PIP5	3

The joint index is transformed via an embedding layer and concatenated with CNN features:

$$
\text{Final Feature} = [\text{CNN(image)} \parallel \text{Joint Embedding}]
$$

3.3 Experimental Design

We conduct four key experiments:

1. **Pooled KL Classifier** (no joint information)
2. **Joint-aware KL Classifier** (with joint index)
3. **Joint Classification Task** (predict joint type)
4. **Per-joint KL Models**

Additionally:

- PCA and UMAP visualization
- ANOVA and Kruskal-Wallis statistical tests

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4. Results

4.1 Overall Performance

Model	Accuracy	Macro F1
Pooled KL	0.7268	0.4666
Pooled + Joint	**0.7466**	**0.5864**
Joint Classifier	**0.9764**	**0.9763**

4.2 Key Findings

1. Joint-aware model significantly improves performance

- Macro F1 increases by **+25%**

This demonstrates that:

Joint position is a critical factor in KL classification.

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2. Joint classifier achieves near-perfect accuracy

- Accuracy: **97.6%**

This indicates:

PIP joints are highly distinguishable based on image features.

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3. Per-joint models underperform

Joint	Macro F1
PIP2	0.407
PIP3	0.352

Joint Macro F1

PIP4 0.321

PIP5 0.481

This suggests:

Splitting the dataset reduces performance due to limited sample size.

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5. Statistical Analysis

5.1 Feature Visualization

(PCA and UMAP plots inserted here)

Observations:

- Clear clustering by joint
- Distinct feature distributions

5.2 Statistical Tests

- ANOVA p-values ≈ 0
- Kruskal-Wallis p-values ≈ 0

Interpretation:

Joint differences are **extremely statistically significant**

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6. Discussion

6.1 Are PIP joints interchangeable?

❌ No.

Evidence:

- Joint classification accuracy = 97.6%
- Strong statistical significance

- Distinct clustering in feature space

6.2 Should we train separate models per joint?

❌ No.

- Performance degrades due to limited data

6.3 Optimal Strategy

✅ **Joint-aware pooled model**

Advantages:

- Uses full dataset
- Preserves joint-specific information
- Achieves best performance

6.4 Clinical Interpretation

Different PIP joints exhibit:

- Structural differences
- Distinct degeneration patterns
- Varying radiographic appearances

Thus:

KL grading must account for anatomical context.

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7. Limitations

- Severe class imbalance (KL3/4)
- Limited samples for advanced disease
- Single-image classification (no multi-joint fusion)

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8. Future Work

8.1 Data Augmentation

Use generative models (CVAE / diffusion) to:

- Augment KL3/4
- Improve severe-class detection

8.2 Multi-stage Classification

- Stage 1: Normal vs Severe
- Stage 2: KL3 vs KL4

8.3 Multi-joint Modeling

Combine multiple joints for:

- Patient-level prediction

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9. Conclusion

This study demonstrates that:

- PIP joints exhibit strong joint-specific differences
- Ignoring joint information significantly reduces performance
- Incorporating joint index improves classification accuracy and F1 score
- The optimal approach is a **joint-aware pooled model**

Final Conclusion

PIP2–PIP5 joints should not be treated as identical inputs.
A joint-aware classification model is essential for accurate KL grading.